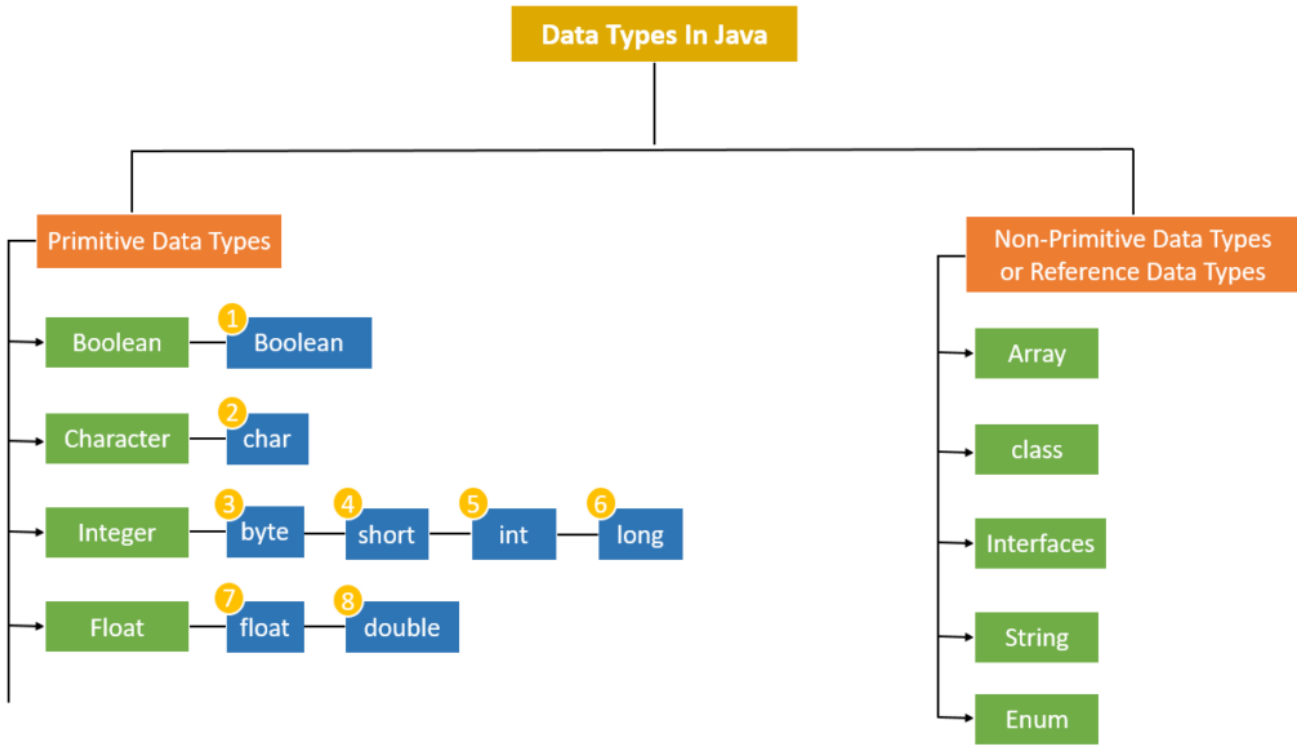
# Experiment 2

## Data Types

### Aim / Objective:

The Java programming language has a rich set of data types. The Java programming language is statically-typed, which means that all variables must first be declared before they can be used. The data type is a category of data stored in variables. The use of data types is crucial for ensuring data integrity and efficient memory usage in Java programs.

### Theory / Concept:




### Problem Statement 1:

**OUTPUT:**

**Problem Statement 2:Program to perform arithmetic operations (add, subtract, multiply, divide).**

**OUTPUT:**

## Compilation and Execution Steps:

Here are the standard steps to **compile and run** a Java program using the command line or terminal:

### Step 1: Write the Java Program

- Create a new text file using any editor (e.g., Notepad, VS Code, or any IDE).
- Save the file with a `.java` extension and ensure the **file name matches the public class name**.

### Step 2: Open Terminal / Command Prompt

- **Windows**: Press `Win + R`, type `cmd`, and press `Enter`
- **Mac/Linux**: Open the Terminal

### Step 3: Navigate to the File Location

- Use the `cd` command to change the directory to where your Java file is saved.
  `cd path\to\your\java\file`

### Step 4: Compile the Java Program

Use the `javac` command (Java compiler) to compile your `.java` file:

```
javac Example.java
```

**Learning Outcomes:**

Students will able to

LO1 Understand Control Statements in java
LO2 Understands the usage of **if**, **if-else**, **if-else-if ladder**, **nested if**, and **switch-case** statements.
LO3 Understands about flow of control in a program based on different logical conditions

**Course Outcome:**

CO1: Apply Different datatypes in programming.

## Conclusion:

## Viva Questions:

1. How many types of data types are there in Java?

2. What is the difference between primitive and non-primitive data types?

3. What is the default value of `int`, `boolean`, and `char`?

## For Faculty Use only:

| Correction Parameters | Formative Assessment [40%] | Timely completion of Practical [ 40%] | Attendance / Learning Attitude [20%] | Total |
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